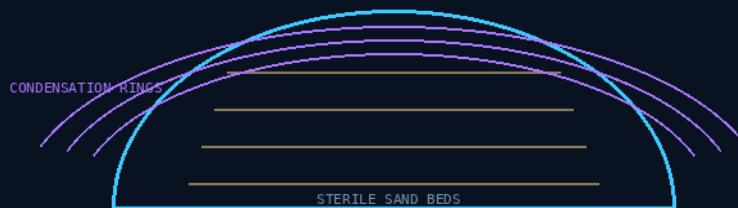


THE STARSHIP GALILEO
DEEP SPACE TRAVEL AND COLONIZATION
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS
PHASE 26

**'SNOWGLOBE' STERILE SAND HYDROPONICS;
VAPOR RECLAMATION CIRCLES**

Sterile sand beds — autotrophic mineral salts — no soil
Space-ice melt to condensate — the sterile closed circle



▼
SPACE-ICE MELT -> DRINK -> TRANSPIRE -> SCAVENGE -> RE-CONDENSE

SOLID WASTE RECYCLING: BANNED — THE HEAVY-METAL LAW
SPECIFIED RECOVERY EFFICIENCY, SIZED THEN CALCULATED — NO ABSOLUTES

GP-IM-026

Revision v1.0 — 20 August 2026

27 of 290

VETTED: PASS ARCHITECTURE — EFFICIENCY TO BE SIZED

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LICENSE: CERN OPEN HARDWARE LICENCE v2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 26

PHASE 26: 'SNOWGLOBE' STERILE SAND HYDROPONICS; VAPOR RECLAMATION CIRCLES — STATUS: CURRENT. The ship's clean farm: solid human waste recycling is BANNED (the heavy-metal law — exponential bioaccumulation is the design enemy); crops grow exclusively in sterile volcanic sand and gravel hydroponic beds fed with autotrophic mineral salts; urine diverts entirely to industrial foam/cooling lines; drinking water is melted harvested space-ice, consumed, transpired, vacuumed out of the cabin humidity, and re-condensed in a sterile closed circle. Vetted: PASS architecture — with the recovery-efficiency discipline seated: no absolute figures; the percentage is sized, then calculated, then seated.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-026, v1.0 — 20 August 2026. The twenty-seventh manual of the program — 27 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the snowglobe law as seated (contents line and design body, verbatim) — §2 why it works (the loop, graded; the efficiency discipline) — §3 the system on the ship — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

VERSION NOTE (v1.0 — 20 August 2026): first build of this manual, on the Author's standing night-shift order (seated verbatim in sitting 499). The phase body stands exactly as seated in the master — split out of the combined PHASE 12, 24, 25 & 26 cluster into its own standalone section in sitting 498, no wording changed; the manual adds the assembly truth around it.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Bypasses vulnerable soil agriculture. Utilizes sterile sand hydroponic beds fed by closed-loop, multi-tier condensation rings that capture and recycle of agricultural moisture vapor with zero loss.' *[VET CONSTRAINT, seated §5: 'zero loss' reads as the design intent, never as a certified figure — the*

defensible specification is a stated recovery efficiency (98%, 99%, 99.9%...), calculated once the system is sized; the author's 15 August 2026 kill order already struck '100%' and 'absolute' from this phase's texts; the line stays preserved verbatim per doctrine.]

THE SEATED DESIGN BODY (verbatim): "'Snowglobe" Life Support: Solid human waste recycling is banned to prevent exponential heavy-metal bioaccumulation. Crops are grown exclusively in sterile volcanic sand and gravel hydroponic beds fed with autotrophic mineral salts. Urine is entirely diverted to industrial foam/cooling lines. Main drinking water is melted from harvested space-ice, consumed, transpired via plants/sweat, vacuumed by cabin humidity scavengers, and re-condensed in a sterile closed circle.'

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE LOOP PHYSICS, AFFIRMED: closed-loop water recovery is completely viable — the vet's own words. Melt, consume, transpire, scavenge, re-condense: every stage is ordinary engineering flown with discipline. The cabin humidity scavengers are the loop's quiet heroes — the farm's transpiration and the crew's sweat both come home as water.

THE HEAVY-METAL LAW, AFFIRMED: solid human waste recycling is banned to prevent exponential heavy-metal bioaccumulation — in a closed ecology, what you eat returns to what you grow; metals concentrate each pass; the ban breaks the spiral at the source. Urine diverts to industrial foam/cooling lines — water value recovered, biology kept out of the food line.

THE EFFICIENCY DISCIPLINE, SEATED AS LAW OF READING: 100% recovery and absolute zero loss are not defensible engineering specifications (the vet, seated). There WILL be residual moisture, dissolved material, plant biomass leaving the loop, leaks, maintenance losses, and unavoidable condensation inefficiency. The correct claim is a SPECIFIED RECOVERY EFFICIENCY — 98%, 99%, 99.9% — calculated once the system is sized. The snowglobe's honesty is its number, owed at §9 until sized.

THE STERILE-SAND LOGIC: volcanic sand and gravel beds fed with autotrophic mineral salts bypass soil's vulnerabilities entirely — no soil-borne pathogens, no soil ecology to manage, no heavy-metal accumulation vector through the growth medium. Sterile in, sterile round.

§3 — THE SYSTEM ON THE SHIP

- THE BEDS: sterile volcanic sand/gravel hydroponic beds, mineral-salt fed — the farm's growth medium is geology, not biology.
- THE BAN: solid waste never re-enters the food loop — banned by design law; the waste line's destinations are industrial (foam/cooling), not agricultural.

- THE WATER: harvested space-ice melted for drinking; consumed; transpired by plants and crew; vacuumed by cabin humidity scavengers; re-condensed — the sterile closed circle.
- THE RINGS: closed-loop multi-tier condensation rings capture the agricultural vapor — the farm's breath comes back as irrigation.

INTEGRATION NOTE (audit, flagged): the farm is mass, water and power — pod placement, ice intake and the scavenger network ride the hull and pod architecture (GP-IM-006 v1.4, GP-IM-019 v1.0); the loop's sized efficiency rides §9.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE TRANSIT TRADE (seated doctrine): the snowglobe matrix 'continuously farm[s] fresh consumer crops inside specialized agricultural pods' — and the trade half: a survey starship cruising past a depot moon at speed swaps 'depleted waste cubes for fresh pre-grown greenhouse food pods' without deceleration burns (Phase 20 hot-swap) — the farm is a currency.
- THE FEEDSTOCK CIRCLE (seated doctrine): harvested vent minerals feed 'the biological food-culture line (Phase 25 mycorrhizal fungi, Phase 26 sand-hydroponics) under ice, where sunlight never reaches' — the farm eats the mining line's output.
- THE CROSS-VET LAW (seated doctrine): new synthesized elements are cross-vetted against 'Phase 25 mycorrhizal fungi, Phase 26 sand-hydroponics, and closed-loop cabin water lines' before joining the ship — the farm is a materials judge.
- THE CREW LINE (seated doctrine): 'Biologists, horticulturalists, medics: 25/26 (fungi and sand hydroponics)...' — the snowglobe owns crew posts.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT, vet batch 11), VERBATIM GRADE: ' PASS architecture / REMOVE "100% / ZERO LOSS".' Its findings, seated: closed-loop water recovery completely viable; but 100% recovery and absolute zero loss 'are not defensible engineering specifications' — residual moisture, dissolved material, biomass leaving the loop, leaks, maintenance losses, condensation inefficiency; 'the correct claim is a specified recovery efficiency, for example 98%, 99%, 99.9%, etc. Once the actual system is sized, we calculate it. So the hydroponic architecture survives. The absolute wording does not.' The author's 15 August 2026 kill-absolutes order had already struck '100%/'/'absolute' from the phase's texts; the remaining 'zero loss' reads as design intent under §2's discipline, the sized figure owed at §9.

§6 — BUILD SEQUENCE

- STEP 1 — Build the beds: sterile volcanic sand/gravel beds with mineral-salt fertigation; sterile handling from fill to harvest.
- STEP 2 — Build the water circle: space-ice intake and melt plant, potable line, humidity scavengers, condensation rings — every stage metered so the sized efficiency can be MEASURED, not asserted (§9).
- STEP 3 — Plumb the diversions: urine to industrial foam/cooling lines; solid waste out of the food loop by hard plumbing law — no valves that could cross the streams.
- STEP 4 — Commission the farm pods: the agricultural pod standard (Phase 19/32 walls, Phase 23 displays, Phase 24 boundary discipline at the farm line).
- STEP 5 — Bench the loop figure (§9): measured recovery efficiency across a full crop cycle — then seat the number in place of the intent.
- STEP 6 — Commission the trade packs: the fresh-food pod format for the Phase 20 hot-swap doctrine — the farm's export line.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 25 — the seeding pods (GP-IM-025): the cluster mate — the fungi line and the farm line feed each other (§4).
- PHASE 12 — the agar bio-siphons (GP-IM-012): the agar and bio-lock discipline the farm's sterile doctrine inherits.
- PHASE 20 — the tugs (GP-IM-020 v1.0): the hot-swap trade the farm's export pods ride (§4).
- PHASE 24 — the scrub boundary (GP-IM-024 v1.0): the farm line's biological border control.
- PHASE 19 — the pods (GP-IM-019 v1.0): the agricultural pod standard; farm furniture on the 19/32 walls.
- PHASE 6 — Hull & Structure (GP-IM-006 v1.4): farm placement, ice intake, the scavenger network's trunking.

§8 — DO-NOT-BUILD

- NEVER recycle solid human waste into the food loop — the heavy-metal law is absolute and seated (§1, §2).
- NEVER claim '100%' or 'zero loss' as a figure — the specified efficiency is sized, measured, seated (§2, §9).

- NEVER cross the diversion plumbing — urine to industry, never to agriculture; hard plumbing law (§6, step 3).
- NEVER let soil aboard the farm line — sterile sand only; soil is the vulnerability the phase exists to bypass (§1).
- NO unmetered stages — what isn't metered can't be certified (§6, step 2).

§9 — OWED

THE BENCH, OWED: the sized recovery-efficiency figure (measured across a full crop cycle — the number that replaces the intent); bed areas and crop calendars per crew head; mineral-salt consumption and the vent-minerals feedstock reconciliation; ice harvest rates vs consumption; scavenger and ring capacities; waste-cube and food-pod formats for the trade line. All owed items ride the bench until spoken.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-026, v1.0 — CURRENT, seated law, master v2.1 (numerical print order, post-split). The twenty-seventh manual of the program — 27 of 290 delivered. Built 20 August 2026.

PROVENANCE — THE STANDING ORDER, VERBATIM (seated in sitting 499): '16 and then bed for me, after we finish that you can do 17 to 27 and i will read all 10 first thing to see if we can get through them faster in blocks'. The phase's own chain, all seated in the master: the design body and Contents line from the July record; the author's 15 August 2026 kill-absolutes order striking '100%/'absolute' from the phase's texts; the naming batches of 12 August 2026 ('26: "Snowglobe" Sterile Sand Hydroponics; Vapor Reclamation Circles'); the vet batch 11 grade with the efficiency discipline; the split of 20 August 2026 that stood the phase alone (sitting 498, his order).

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete. Next version triggers: the sized efficiency figure, the §9 benches, or his word.

END OF MANUAL — PHASE 26